

NANCY-ESSEY AP, France

WMO: 071800

Lat: 48.6878N

Lon: 6.2214E

Elev: 229

StdP: 98.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.8	-11.3	1.5	-6.7	-8.9	1.8	-4.7	10.7	7.9	9.6	7.7	2.1	40	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.7	20.1	29.5	19.5	27.6	18.9	21.2	29.0	20.4	27.5	19.6	25.9	3.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.0	23.7	18.0	13.3	22.8	17.2	12.6	22.2	62.5	28.9	59.5	27.5	56.7	26.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.5	-11.4	34.7	3.0	2.2	-13.5	36.3	-15.3	37.6	-17.0	38.8	-19.2	40.4		
(5)			-11.8	23.0	2.9	0.9	-13.9	23.6	-15.6	24.1	-17.3	24.6	-19.4	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	2.4	3.5	6.6	9.9	14.0	17.7	19.8	19.3	15.2	11.3	6.4	3.3		
	DBStd	7.13	4.51	4.14	3.67	3.50	3.39	3.47	3.18	3.22	3.02	3.67	3.85	4.32		
	HDD10.0	938	236	183	113	44	5	0	0	0	1	30	118	208		
	HDD18.3	2907	493	415	362	252	139	54	21	27	102	220	359	464		
	CDD10.0	1244	1	1	9	42	131	231	303	288	157	69	9	2		
	CDD18.3	171	0	0	0	0	6	34	66	56	8	1	0	0		
	CDH23.3	1872	0	0	0	10	91	389	708	571	98	4	0	0		
	CDH26.7	617	0	0	0	0	17	119	255	208	18	0	0	0		
(14) Wind	WSAvg	2.9	3.2	3.3	3.2	3.0	2.8	2.8	2.7	2.4	2.5	2.6	2.8	3.2		
(15) Precipitation	PrecAvg	768	62	58	58	51	68	69	64	66	65	63	70	74		
	PrecMax	983	154	159	153	137	198	179	172	140	134	189	185	201		
	PrecMin	469	11	8	5	10	9	7	8	4	6	2	14	4		
	PrecStd	131	33	39	34	32	31	39	33	36	36	40	37	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.2	20.2	24.7	28.9	33.2	34.9	35.6	28.9	23.9	17.2	13.8		
		MCWB	11.0	9.5	12.2	15.5	19.3	20.6	21.0	20.6	18.7	16.8	13.4	10.9		
	2%	DB	11.6	12.8	17.6	22.3	25.7	30.0	31.6	31.3	26.2	21.1	14.9	12.2		
		MCWB	9.6	8.9	11.1	14.2	17.3	19.9	20.2	20.2	18.0	16.1	12.2	10.2		
	5%	DB	10.1	11.1	15.1	19.7	23.6	27.6	29.5	28.7	23.7	18.9	13.4	10.9		
		MCWB	8.4	8.3	10.1	12.9	16.3	19.2	19.5	19.2	17.1	15.1	11.4	9.3		
	10%	DB	8.7	9.5	12.9	17.4	21.4	25.4	27.4	26.5	21.5	16.9	11.9	9.5		
		MCWB	7.1	7.4	9.3	11.8	15.2	18.1	18.7	18.6	16.1	14.1	10.3	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.5	13.2	16.6	20.0	22.1	22.4	22.2	20.2	17.9	14.2	11.7		
		MCDB	12.7	13.4	18.1	23.4	26.9	29.8	30.9	30.4	26.1	22.0	16.2	12.9		
	2%	WB	10.0	9.6	11.8	14.7	18.4	20.8	21.2	21.0	18.9	16.6	12.6	10.5		
		MCDB	11.3	12.0	15.7	20.9	24.4	27.6	29.3	29.0	24.7	19.9	14.4	12.0		
	5%	WB	8.4	8.6	10.8	13.3	17.0	19.7	20.2	20.2	17.8	15.4	11.6	9.3		
		MCDB	9.9	10.7	14.3	18.6	22.2	26.1	27.6	26.9	22.6	18.2	13.2	10.8		
	10%	WB	7.2	7.5	9.7	12.2	15.8	18.8	19.3	19.2	16.8	14.4	10.4	8.2		
		MCDB	8.6	9.4	12.7	16.6	20.3	24.6	25.9	25.0	20.5	16.8	11.8	9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.9	9.3	10.9	10.9	11.4	11.2	11.2	10.5	8.7	5.9	5.0		
		MCDBR	6.4	9.9	14.2	16.0	15.0	15.5	15.6	16.0	14.6	11.7	7.9	6.2		
	5% WB	MCWBR	5.0	6.8	8.6	8.9	7.6	7.0	5.9	6.5	7.3	7.1	5.6	4.8		
		MCWBR	6.0	8.4	11.6	14.1	13.6	14.2	14.0	14.1	12.6	10.3	7.3	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.335	0.362	0.392	0.402	0.407	0.402	0.399	0.373	0.358	0.328	0.308		
		taud	2.437	2.384	2.321	2.255	2.265	2.284	2.290	2.325	2.372	2.405	2.434	2.421		
	Ebn at Noon		754	814	847	855	859	855	853	839	826	774	722	703		
		Edh at Noon	65	85	106	125	129	128	125	116	100	83	65	58		
(44) All-Sky Solar Radiation	RadAvg		0.90	1.65	2.87	4.30	5.08	5.79	5.43	4.69	3.59	2.12	1.04	0.72		
	RadStd		0.14	0.29	0.43	0.57	0.57	0.50	0.53	0.49	0.36	0.17	0.10	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page